

Warmup/Attendance: [Week 4: Friday 9/18/2026]

Full Name:

SID:

- As you sit down in class, start working with the people around you to try to solve these problems
- These are *not* graded for correctness, but you will turn this sheet in for attendance at the end of class.
- We'll discuss the solutions in lecture today.

Equivalent or Different commands?

Are the following commands equivalent (i.e. give you the same plot)?

1.

```
# command 1.1
ggplot(iris, aes(x = Petal.Length, y = Petal.Width, color = Species)) +
  geom_point()

# command 1.2
ggplot(iris, aes(x = Petal.Length, y = Petal.Width)) +
  geom_point(aes(color = Species))
```

2.

```
# command 2.1
ggplot(iris, aes(Petal.Length, Petal.Width)) +
  geom_point(color = "blue")

# command 2.2
ggplot(iris, aes(Petal.Length, Petal.Width, color = "blue")) +
  geom_point()
```

3.

```
# command 3.1
ggplot(iris, aes(Petal.Length, Petal.Width)) +
  geom_point()

# command 3.2
ggplot(iris) +
  geom_point(aes(Petal.Length, Petal.Width))
```

What is the issue in these commands?

Can you spot the bug?

4.

```
ggplot(iris) +  
  geom_point(x = Petal.Width, y = Petal.Length)
```

5.

```
ggplot(iris, aes(Petal.Length, Petal.Width))  
  geom_point(aes(color = Species))
```

6.

```
ggplot(iris, aes(x = Petal.Length, y = Petal.Width)) |>  
  geom_point(aes(color = Species))
```